

## Internships & Courses recommendation system

### Project overview:

The core objective is to build an automated system that:

- **Classifies** the best-matching internship program based on the user's entered **(job title or major)**, from our current 8 programs (Software, Quality, OPD, Physiotherapy, PMO, Dentistry, HR, and Pharma Internship), if there's no matching, it will return **"Not matched"**.
- Uses a **Large Language Model (LLM)** which is **(Lama 4 maverick)** to recommend the most relevant **course** from our existing course list in Andalusia Academy.
- If no match is found, the model will return **"Not matched"**.

### Project workflow:

- The user submits their information via a form (similar to the one shown in the attached image), including details like name, education level, and major.
- The model extracts the job title and performs:
  - 1- Internship program classification.
  - 2- Course recommendation using the LLM.
- We will collaborate with the BI team to deploy the model on the backend of incoming lead data (e.g., from social media ads), storing the results in a pivot table to be visualized in the CRM or dashboard.

### Project implementation:

- The model works on the data to be found in the pivot table and CRM, as we mentioned before.
- A publishing social media campaign that invites users to enter their job title and optionally their skills to improve. The LLM then provides a tailored course recommendation.
- The model prompt will be customized based on the fields used (job title only or job title + skills).

You will find below the link of the model demo: <http://10.24.115.190:8501>

Continued,

## Project demo:

### 1- Sample input: Ex1

Name

Ziad Mohammad Saad

Date of birth ?

2002/01/06

Education status

Graduated

Select your major

Law

Job Title

interested in web development

Select the internship program

Software

Submit

### 2- Sample output: Ex1

☒ Matched

Matched Course: Full Stack Web Development

### 1- Sample input: Ex2

Name

#### #### ####

Date of birth ?

1999/06/17

Education status

Graduated ▼

Select your major

Physiotherapy ▼

Job Title

sports injury specialist

Select the internship program

opd lead ▼

Submit

### 2- Sample output: Ex2

✗ Not matched, but matched with **Physiotherapy**

Matched Course: ATLS (Advanced Trauma Life Support)

### Cost analysis:

In our project, we use an advanced paid AI language model to recommend the most suitable training course based on job titles. This model charges based on how much text we send to it (input) and how much text it returns (output), rather than per request.

### What Are Tokens?

A token is a small chunk of text. It could be as short as a single character or as long as a word. For example:

- AI is very smart → 4 tokens (one for each word)

So, longer texts use more tokens. Every time we send a job title to the model, it uses:

- **Input tokens:** to process the text we send (like the prompt and job title).
- **Output tokens:** to generate a response (e.g., the recommended course).

### Cost calculations:

The pricing of the model we're using (LLaMA 4 Maverick) is:

- **\$0.22 per million input tokens**
- **\$0.88 per million output tokens**

In our testing:

- A typical request uses **around 520 input tokens**
- The output is usually **around 8–10 tokens**
- For 10 requests, our cost was **less than \$0.01**

Based on these numbers, **with a \$50 subscription plan** (Which is the lowest plan), we could make:

- **Up to almost 41120 recommendations or trials or submissions per subscription** before hitting the token cost limits.